Reading a protein across scales

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Abstract

A protein can be described at many scales at once: as a sequence with a confidence profile, as a shape in three dimensions, as residues that differ in how buried and how conserved they are, as a structure that breathes, as a catalyst with measurable kinetics, and as a trajectory through time. Each scale has its own measurement, and each measurement has its own way of being misread. This primer walks through one figure whose six panels show six of these views side by side. We begin with the confidence that AlphaFold assigns to each residue of its model of the tumour suppressor p53, where a folded DNA-binding domain stands out against long, disordered tails. We then show the spike glycoprotein of SARS-CoV-2 as a ribbon and as a surface, and explain what each rendering hides. Four panels use simulated data, generated to show textbook behaviour rather than to report new results: the relation between burial, conservation and crystallographic B-factors; hydrogen–deuterium exchange from the surface of a protein to its core; the signatures of three classes of enzyme inhibitor; and eight simulated molecules as they fold. For each panel we state what the axes mean, what a reader can safely conclude, and what they cannot.

Introduction

Figures in structural biology are dense. A single figure may combine a prediction, an experimental structure, a statistical summary and a simulation, each drawn in the conventions of its own field. Readers from neighbouring fields often know one of these conventions well and the others only by sight.

Here we use one six-panel figure (Figure 1) as a worked example. Panels a and b are drawn from public data: a model from the AlphaFold Protein Structure Database (Varadi et al. 2024) and a cryo-EM structure from the Protein Data Bank (Berman et al. 2000). The remaining panels (Figure 1c–f) are simulations with known parameters, chosen so that the correct reading is not in doubt. We treat them as exercises, not as findings.

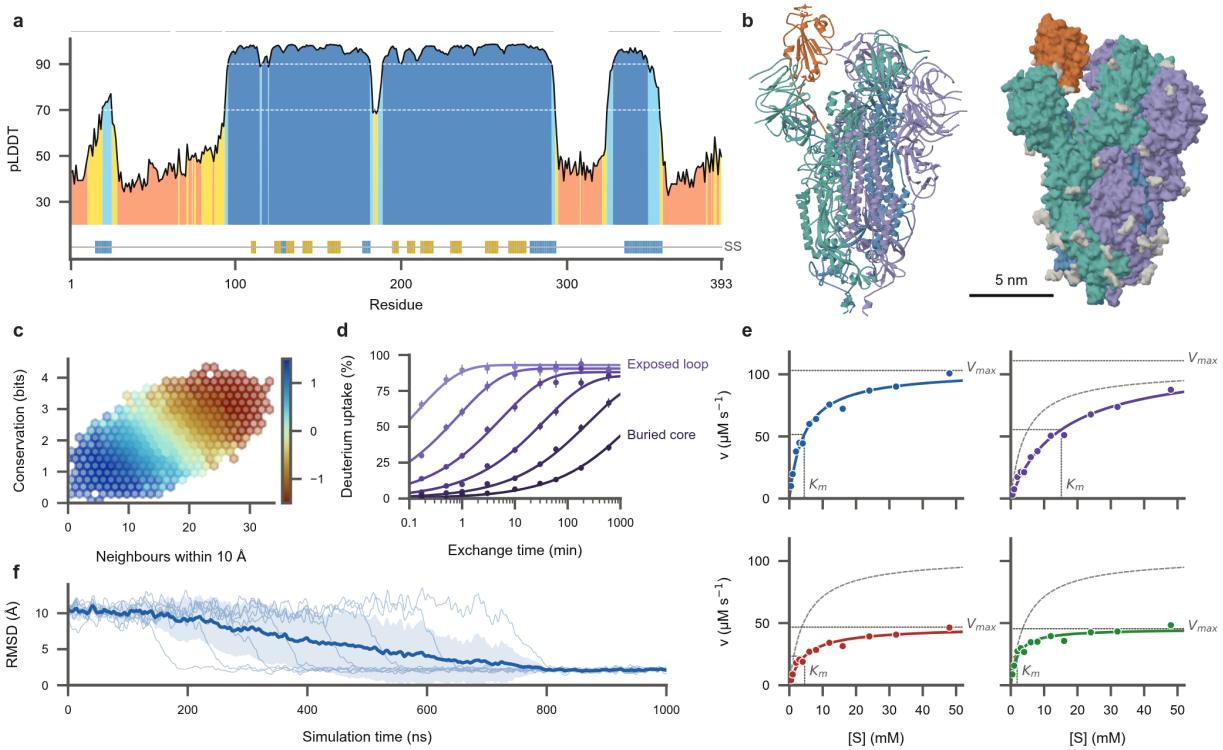


Figure 1. Six views of proteins, from sequence to dynamics. Panels a and b show public data; panels c–f show simulated data made to illustrate typical behaviour. **a**, AlphaFold confidence (pLDDT) along the 393 residues of human p53, shaded by confidence band, with its domains above and secondary structure below. **b**, The SARS-CoV-2 spike trimer (PDB 6VSB) as a ribbon and as a molecular surface; the raised receptor-binding domain is orange. Scale bar, 5 nm. **c**, Mean B-factor of 60,000 residues binned by burial and conservation; opacity shows how many residues fall in each hexagon. **d**, Deuterium uptake of six peptides, from an exposed loop to the buried core, with stretched-exponential fits (points, mean ± s.d.). **e**, Michaelis–Menten kinetics without inhibitor and with competitive, non-competitive and uncompetitive inhibition (row by row from top left); dashed, the uninhibited curve. **f**, Cα RMSD from the native structure in eight folding simulations (thin lines), with their mean ± s.d.

Results

Confidence along the sequence

For every residue, AlphaFold reports a predicted local distance difference test score, pLDDT, between 0 and 100 (Jumper et al. 2021). It estimates how well the model would agree with an experimental structure under the lDDT measure (Mariani et al. 2013). For human p53 the 393-residue profile is far from uniform (Figure 1a). The DNA-binding domain (residues 94–292) and the short tetramerisation helix score above 90, where the backbone and most side chains are expected to be accurate. The transactivation domain, the proline-rich region and the C-terminal domain fall

to between 30 and 50.

Low confidence here is not a failure of the model. These regions are intrinsically disordered in solution (Wright and Dyson 1999; Joerger and Fersht 2008), and very low pLDDT predicts disorder well across the human proteome (Tunyasuvunakool et al. 2021; Ruff and Pappu 2021). About 40% of the residues of p53 score below 70. They should be read as flexible, not as wrong.

Two renderings of one structure

The prefusion spike trimer (Wrapp et al. 2020), with one receptor-binding domain raised, is shown twice in Figure 1b. The ribbon traces the backbone, which makes the secondary structure and the raised domain easy to follow. The surface shows what a binding partner meets: a closed envelope dotted with the first sugars of the glycans. Neither is more correct. The ribbon hides how crowded the molecule is, and the surface hides the fold.

Burial, conservation and mobility

The hexagonal map in Figure 1c summarises 60,000 simulated residues by how buried each one is, counted as neighbouring residues within 10 Å, and how conserved it is, in bits of information in its alignment column (Capra and Singh 2007). Each hexagon is coloured by the mean B-factor of its residues, so buried, conserved residues should be the least mobile. They are, but the map alone cannot say which property matters: burial and conservation are themselves correlated, and only a model that separates them can apportion the effect (Franzosa and Xia 2009). Real B-factors also carry crystal contacts, resolution and refinement, which is why they are normalised within each structure before they are pooled (Sun et al. 2019).

Exchange from surface to core

Hydrogen–deuterium exchange reports how quickly backbone amide hydrogens trade places with deuterium from the solvent (Masson et al. 2019). Six simulated peptides span more than three orders of magnitude in rate (Figure 1d). The exposed loop is nearly saturated within a minute, while the peptide from the buried core has exchanged about a third of its amides after ten hours. Slow uptake signals protection by hydrogen bonds and burial, not slow chemistry: unstructured peptides exchange orders of magnitude faster, at rates that can be predicted from sequence (Bai et al. 1993).

Three ways to inhibit an enzyme

Each plot shows twelve simulated initial rates against substrate concentration, a Michaelis–Menten fit (Johnson and Goody 2011) and, dashed, the uninhibited curve (Figure 1e). Without inhibitor, V_{max} is 100 μM s^{-1} and K_m is 4 mM. A competitive inhibitor raises the apparent K_m to 12 mM and leaves V_{max} unchanged, because enough substrate outcompetes it. A non-competitive inhibitor lowers V_{max} to 45 μM s^{-1} and leaves K_m alone. An uncompetitive inhibitor binds only the enzyme–substrate complex and lowers both by the same factor, so the initial slope at low substrate does not change.

Folding in silico

The last panel follows the Cα root-mean-square deviation from the native structure, after optimal superposition (Kabsch 1976), in eight simulated folding runs of 1 µs (Figure 1f). Each run starts unfolded near 10 Å and drops to about 2 Å at its own moment, between roughly 150 and 760 ns. The mean declines smoothly, and its band is widest in the middle: the average describes none of the runs. Long simulations of small proteins look like this, long waits broken by fast transitions (Lindorff-Larsen et al. 2011), which is why a folding rate is estimated from many runs rather than one.

Methods

Public data. The p53 model (UniProt P04637) and its pLDDT values were taken from the AlphaFold Protein Structure Database (Varadi et al. 2024). The spike is PDB entry 6VSB (Wrapp et al. 2020), drawn as cartoon and molecular surface with open-source PyMOL. Domain boundaries for p53 follow Joerger and Fersht (Joerger and Fersht 2008).

Simulated data. Panels c–f were generated from stated models with fixed random seeds. Panel c: burial and conservation drawn from a bivariate normal distribution (r = 0.55), with B-factor z-scores falling linearly in both, plus Gaussian noise; hexagons with fewer than four residues are left empty. Panel d: stretched-exponential uptake with rates from 8 to 6 × 10^{-4} min^{-1}. Panel e: Michaelis–Menten rates with 4.5% proportional and 1.2 µM s^{-1} additive noise, fitted by non-linear least squares. Panel f: logistic folding transitions with correlated noise.

Figure. Plots were drawn with fluxplot and composed in Flux.

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